

AI COURSE

FROM COLLEGE TO IT INDUSTRIES
CONCEPTS

ARTIFICIAL
INTELLIGENCE



Published By

TSAR IT PRIVATE LIMITED

CONCEPTS

- 1.INTRODUCTION TO AI
- 2.PYTHON FOR AI
- 3.MATHEMATICS FOR AI
- 4.MACHINE LEARNING (ML)
- 5.DEEP LEARNING
- 6.NATURAL LANGUAGE PROCESSING (NLP)
7. COMPUTER VISION
- 8.AI TOOLS & FRAMEWORKS
- 9.INDUSTRIAL APPLICATIONS
- 10.PROJECTS & HANDS-ON
11. INTERVIEW PREPARATION



ABOUT OUR COMPANY

At **TSAR IT Private Limited**, we are a cutting-edge software company dedicated to providing innovative digital solutions to businesses and organizations. Located in Andhra Pradesh, we specialize in a range of services and software products tailored to meet the unique demands of our clients.

we like challenges to solve and give solutions to your business growth and you can monitor your business and employees work easily,

We offer services for a wide range of fields such as software development, web development, graphic development, cross platform development etc. For more information navigate to our services section.

“Explore The Services We Offer For You” we are experts in these services, we have strong experience and technical expert team, we have our own German Technology server, our team will complete and give you project in time.

we will provide our services like how clients feel comfortable, we can handle any type of project with low cost and we are expert in install and maintaining projects.

N RAKESH - CEO

Internship Courses:

In addition to our product offerings, **TSAR IT Private Limited** also provides Internship Courses for aspiring software developers, engineers, and tech enthusiasts. Our internship programs offer hands-on experience and training in cutting-edge technologies, including:

- **Java Development**
- **Spring Boot & React Full-Stack Development**
- **AI & Machine Learning**
- **Cloud Computing**
- **Software Testing & QA**

These internships are designed to bridge the gap between academic learning and real-world software development, preparing the next generation of IT professionals.

At **TSAR IT Private Limited**, we believe in the power of technology to drive success, innovation, and transformation. With a dedicated team of experts, we are committed to delivering high-quality, customized software solutions while fostering new talent through our internship programs.

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ARTIFICIAL INTELLIGENCE SYLLABUS

1. INTRODUCTION TO AI

- 1.1. What is Artificial Intelligence?
- 1.2. History and Evolution of AI
- 1.3. Types of AI: Narrow, General, Superintelligent
- 1.4. Applications of AI in Industries
- 1.5. Ethical Considerations in AI

2. PYTHON FOR AI

- 2.1. Python Basics for AI
- 2.2. NumPy, Pandas, and Matplotlib
- 2.3. Object-Oriented Programming in Python
- 2.4. File Handling and Data Manipulation

3. MATHEMATICS FOR AI

- 3.1. Linear Algebra
- 3.2. Probability and Statistics
- 3.3. Calculus for Machine Learning
- 3.4. Optimization Techniques

4. MACHINE LEARNING (ML)

- 4.1. Introduction to Machine Learning
- 4.2. Supervised vs. Unsupervised Learning
- 4.3. Regression: Linear, Polynomial
- 4.4. Classification: Logistic Regression, SVM, Decision Trees
- 4.5. Clustering: K-Means, Hierarchical
- 4.6. Model Evaluation: Accuracy, Precision, Recall, F1-Score

5. DEEP LEARNING

- 5.1. Neural Networks Basics
- 5.2. Activation Functions
- 5.3. Backpropagation and Gradient Descent
- 5.4. Convolutional Neural Networks (CNNs)
- 5.5. Recurrent Neural Networks (RNNs)
- 5.6. Transformers and Attention Mechanisms

6. NATURAL LANGUAGE PROCESSING (NLP)

- 6.1. Text Preprocessing
- 6.2. Tokenization, Stemming, Lemmatization
- 6.3. Word Embeddings: Word2Vec, GloVe
- 6.4. Sentiment Analysis
- 6.5. Chatbots and Language Models (e.g., GPT, BERT)

7. COMPUTER VISION

- 7.1. Image Processing Basics
- 7.2. Feature Detection and Extraction
- 7.3. Object Detection (YOLO, Faster R-CNN)
- 7.4. Image Segmentation

8. AI TOOLS & FRAMEWORKS

- 8.1. TensorFlow and Keras
- 8.2. PyTorch
- 8.3. Scikit-learn
- 8.4. OpenCV
- 8.5. Hugging Face for NLP

9. INDUSTRIAL APPLICATIONS

- 9.1. AI in Healthcare
- 9.2. AI in Finance
- 9.3. AI in Autonomous Vehicles
- 9.4. AI in E-commerce

10. PROJECTS & HANDS-ON

- 10.1. Predictive Modeling
- 10.2. Image Classification
- 10.3. Sentiment Analysis
- 10.4. AI Chatbot Development
- 10.5. Deploying AI Models (Flask/Django)

11. INTERVIEW PREPARATION

- 11.1. Common AI Interview Questions
- 11.2. Case Studies
- 11.3. Mock Interviews

For Registration process:

Step 1:

Open your Browser & paste the link given below

Step 2:

Given Your Details

Step 3:

Click Register Button

REGISTRATION LINK :<https://internship.tsaritservices.com/registrationForm>

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For More Details:

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